

Final Portfolio

Sharlize Berrospe

Math 122

Complex Variables and Applications

December 16, 2024

I REFLECT: #1 - UNIT 2 - PROOF TECHNIQUES

ϵ - δ DEFINITION FOR A COMPLEX VARIABLE

$$\lim_{z \rightarrow z_0} f(z) = W_0 \iff \forall \epsilon > 0 \exists \delta > 0 \text{ s.t. if } 0 < |z - z_0| < \delta \text{ then } |f(z) - W_0| < \epsilon$$

1. Use the ϵ - δ definition of a complex limit to prove the following statement: $\lim_{z \rightarrow z_0} (z^2 + c) = z_0^2 + c$

In this problem we are given the limit of $|z^2 + c|$ as z approaches z_0 and expected to prove that the limit is equivalent to $z_0^2 + c$.

2. I chose this problem since it was one from discussion 05 which I was confused about for a while, and did not know how to approach the scratchwork / proof parts.

Scratchwork:

Let $\epsilon > 0$

Need to find $\delta > 0$ such that: initially, after this step I substituted $z = x + iy$ and $z_0 = x_0 + iy_0$ } I didn't realize it was completely unnecessary to do so

$$|(z^2 + c) - (z_0^2 + c)| < \epsilon \rightarrow |z^2 - z_0^2| < \epsilon \rightarrow |(z - z_0)(z + z_0)| < \epsilon$$

$$|f(z) - W_0| < \epsilon; f(z) = z^2 + c, W_0 = z_0^2 + c$$

3. A large part that helped me understand where to go from $\uparrow$ this step was the "scam" we were taught in lecture. By knowing it was possible to add/subtract a certain term, I then understood where to go for the rest of the problem. Also really understanding how the triangle inequality fits into this problem and how knowing that could simplify this problem much more for me.

$$\rightarrow |z - z_0| |z + z_0| < \epsilon \rightarrow \text{proof tip "scam": add/subtract a term you want or don't}$$

$$|z - z_0| |z - z_0 + 2z_0| < \epsilon$$

$$|z - z_0| < \frac{\epsilon}{1 + 2|z_0|}$$

assume

$$\begin{cases} \delta < 1 \\ |z - z_0| < \delta < 1 \end{cases}$$

assume

$$0 < |z - z_0| < \delta$$

$$|z - z_0| (1 + 2|z_0|) < \epsilon$$

triangle inequality:

$$|z + z_0| = |z - z_0 + 2z_0| \leq |z - z_0| + 2|z_0| < 1 + 2|z_0|$$

$$\text{goal: } |z - z_0| < \delta \Rightarrow |z^2 - z_0^2| < \epsilon$$

$$\underbrace{|z - z_0|}_{< \delta} \underbrace{|z + z_0|}_{\text{problem}} + |z - z_0| 2|z_0| \leq \underbrace{|z - z_0|}_{< \delta} + \underbrace{2|z_0|}_{?}$$

this was another part of this problem and

$$|z^2 - z_0^2| < \delta (8 + 2|z_0|)$$

many others like it that had stumped me since I had trouble understanding why we just assumed something to be true. I felt like it was incorrect/improper to do so, but I eventually learned it was a necessary step to make that assumption.

Proof

Given $\varepsilon > 0$, choose δ . What if $\delta \leq 1$ or $\delta \leq \frac{\varepsilon}{1+2|z_0|}$?

$$|z^2 + c - (z_0^2 + c)| \leq \delta(\delta + 2|z_0|) \rightarrow |z^2 + c - (z_0^2 + c)| \leq \delta(\delta + 2|z_0|) \\ < \delta(1 + 2|z_0|) \\ \text{let } \delta = \min \left\{ 1, \frac{\varepsilon}{1+2|z_0|} \right\} \quad < \frac{\varepsilon}{1+2|z_0|} (1+2|z_0|)$$

$$|z^2 + c - (z_0^2 + c)| < \varepsilon \Rightarrow \text{Therefore, } \lim_{z \rightarrow z_0} z^2 + c = z_0^2 + c \quad \checkmark$$

#2 - UNIT 7: RESIDUES AND POLES

- 1) Find the Laurent Series representation for the function $f(z) = z^2 \sin\left(\frac{1}{z}\right)$ in the domain $0 < |z| < \infty$.

In this problem from discussion we were provided with a list of relevant Taylor Series of varying functions and the domains in which they were analytic. The Taylor Series most relevant to this problem was the following: $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$; $|z| < \infty$. We were also provided with the Laurent Series definition and series representation for $f(z)$ at each point in the domain.

- 2) My initial interpretation heavily relied on the definition of a Laurent Series of $f(z)$, a_n , and b_n .

Despite doing similar problems as a class with the TA, I interpreted needing to find the Laurent Series representation of $f(z)$ as utilizing the Taylor Series of $\sin z$, then solving for

$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=0}^{\infty} \frac{b_n}{(z-z_0)^n}$; $0 < |z| < \infty$. Before realizing my error and overcomplicating the problem, I attempted to use what I first found (the actual right answer), and plug it into the Laurent series where I thought it should go.

- 3) To realize my mistake, after class I crosschecked the lecture notes and problems the TA did with us to fill in my blanks and answer my own questions. It was then that I had realized I made the problem more difficult than necessary only because it seemed too simple to be right - and I felt as though I was missing something when I wasn't.

4) $f(z) = z^2 \sin\left(\frac{1}{z}\right) = z^2 \sin\left(z^{-1}\right) = z^2 \sum_{n=0}^{\infty} (-1)^n \frac{z^{-2n+1}}{(2n+1)!} = z^2 \sum_{n=0}^{\infty} (-1)^n \frac{z^{-4n+2}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{-4n}}{(2n+1)!}$ Actual answer

given $\uparrow$ simplified $\uparrow$

$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$

$\frac{z^{-4n-2}}{z} = \frac{z^{-4n-2+2}}{z} = \frac{z^{-4n}}{z}$

Laurent series for $f(z) = z^2 \sin\left(\frac{1}{z}\right)$ is $f(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{-4n}}{(2n+1)!}$ in the domain

$0 < |z| < \infty$, using the Taylor Series for $\sin z$: $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$; $|z| < \infty$.

#3 - UNIT 3: ANALYTIC FUNCTIONS

1) Let $f(z) = \operatorname{Re}(z)$ and $g(z) = \operatorname{Im}(z)$.

(a) prove that $f'(z)$ dne at any point

(b) prove that $g'(z)$ dne at any point

(c) Let $h(z) = f(z) + g(z)$. Prove that $h'(z)$ exists at every point

for this problem, we must disprove (a and b) and prove (c) differentiability using the cauchy-riemann equations and the real or imaginary parts of z . We used our knowledge of knowing that for $f'(z)$ to exist at a point, the first-order partial derivatives of u and v must exist at (x_0, y_0) , or $u_x = v_y$ AND $u_y = -v_x$ must be satisfied.

2) (a) $z = x + iy$; $f(z) = u(x, y) + iv(x, y)$

And $f(z) = \operatorname{Re}(z) = x + i(0) = x$

3) $u_x = v_y$ $u_y = -v_x$
 not satisfied $1 \neq 0$ satisfied $0 \neq 0$

In order for $f'(z)$ to not exist at any point, at least one cauchy-riemann equation must not pass. Since one doesn't, $f'(z)$ dne at any point.

parts a) and c) I initially was not confused about and understood how to do.

(b) $z = x + iy$; $g(z) = u(x, y) + iv(x, y)$

$g(z) = \operatorname{Im}(z) = i[0 + iy] = iy = iy$

$u_x = v_y$ $u_y = -v_x$
 not satisfied $0 \neq 1$ satisfied $0 \neq 0$

In order for $g'(z)$ to not exist at any point, at least one cauchy-riemann equation must not pass. Since one doesn't, $g'(z)$ dne at any point.

When I first completed this problem, I had wrote that $g(z) = \operatorname{Im}(z) = -y$ because I believed that $\operatorname{Im}(z) = 0 + iy = iy$.

I did not realize that when accounting for only the imaginary portion of $z = x + iy$, we are considering y itself only, not the imaginary value i . Therefore, I wrote

$g(z) = i(iy) = (-1)y = -y$, when it should have been $g(z) = i(y) = iy$, which is important for part (c).

4) (c) $h(z) = f(z) + g(z) = (x + i(0)) + (i(0) + i(y)) = x + i(y) = x + iy$

$h(z) = x + iy = z$
 $u_x = v_y$ $u_y = -v_x$
 $1 = 1$ $0 = 0$
 Satisfied Satisfied

initially, I did not apply the C.R. equations as I did not realize they were needed for the proof

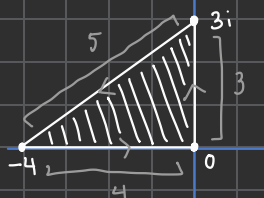
In order for $f'(z)$ to exist at any point, both cauchy-riemann equations must pass. Since both do pass, $h'(z)$ exists at any point.

In the corrected format, we proved with the partials of $h(z)$ and C.R. equations that $h'(z)$ does exist at every point.

just like before. I also believed it not necessary to prove $h(z) = x + iy = z$ + $h'(z) = \frac{d}{dz}[z] = 1$, which is not a proof and not correct. Taking the derivative of $h(z)$ as if it is a simple function and not complex is improper.

#4 - UNIT 5 · COMPLEX INTEGRATION

- 1) Show that if C is the boundary of the triangle with vertices at the points $0, 3i$, and -4 ,
- 2) oriented in the ccw direction, then $\left| \int_C (e^z - \bar{z}) dz \right| \leq 60$
- 3)
- 4)



$$L = 5 + 4 + 3 = 12$$

$$L = 12$$

$$\left| \int_C (e^z - \bar{z}) dz \right| \leq 60$$

$$\left| \int_C f(z) dz \right| \leq M \cdot L$$

$L =$ one length of C

$$\left| \int_C (e^z - \bar{z}) dz \right|$$

$$\downarrow f(z) = e^z - \bar{z}$$

We know $|z_1 - z_2| \leq |z_1| + |z_2|$

$$|e^z - \bar{z}| \leq |e^z| + |\bar{z}|$$

$$\begin{aligned} \text{LHS } z = x + iy \\ e^z &= e^{x+iy} = e^x e^{iy} \\ &= e^x (\cos y + i \sin y) = |e^x| (\cos y + i \sin y) \\ &= e^x |\cos y + i \sin y| = e^x \sqrt{(\cos y)^2 + (\sin y)^2} \\ &= e^x \sqrt{\cos^2 y + \sin^2 y} = e^x \sqrt{1} \\ &= e^x \end{aligned}$$

$$\text{RHS } \bar{z} = x - iy$$

$$|\bar{z}| = |x - iy| = \sqrt{x^2 + y^2}$$

$$|e^z| + |\bar{z}| = |e^x| + \sqrt{x^2 + y^2} = e^x + \sqrt{x^2 + y^2} = 5$$

$x=0, y=3 \Rightarrow e^0 + \sqrt{0^2 + 3^2} = 1 + 3 = 4$
 $x=-4, y=0 \Rightarrow e^{-4} + \sqrt{(-4)^2 + 0^2} = e^{-4} + 4 \approx 4.14$
 $M = 5$

$$\left| \int_C (e^z - \bar{z}) dz \right| \leq \underbrace{M \cdot L}_{5 \cdot 12} = 60$$

$$5 \cdot 12 = 60 \quad \checkmark$$

When we worked on this problem

in discussion, I was confused on how we got from the LHS to the RHS of this. I did not understand

that we were utilizing

$$|z_1 - z_2| \leq |z_1| + |z_2|$$

This confusion and lack of understanding caused me to be confuse later down

the problem for a time.
 $z_1 = e^z$
 $z_2 = \bar{z}$

Another portion of the problem I was lost on was where M , the max, was derived from.

I had trouble grasping the notion that

$M=5$ referred to the max value of z in the complex plane C for the given function $f(z)$. I also realized we are finding this max with the distance from the distance of the origin to each vertices provided.

2 EVALUATE

UNIT 1: (1) A boundary point ($z_0 \in \mathbb{C}$) in a set ($S \subseteq \mathbb{C}$) may or may not also be an accumulation point.

ANSWER: TRUE

TRUE: Whether or not a boundary is also considered an accumulation point depends on if every deleted neighborhood of z_0 intersects S . For z_0 to be a boundary point, z_0 is neither an exterior/interior point. For z_0 to be an accumulation point, every deleted neighborhood (neighborhood w/o z_0) of z_0 intersects S . This can seem almost contradictory to someone w/o knowledge on the topic.

UNIT 4: (2) Satisfaction of the Cauchy-Riemann equations guarantees a complex function $f(z)$ is "entire" everywhere.

ANSWER: FALSE

FALSE: To be entire, a function must be analytic everywhere. In other words, $f(z)$ is analytic at a point z_0 if it is differentiable (Cauchy-Riemann eqs. pass) at every point in some neighborhood around z_0 (continuity). Cauchy-Riemann eqs. passing at a point z_0 , and the partial derivatives being continuous imply differentiability. On the same vein, if a function is differentiable it is implied that the CR eqs. pass and the partials are continuous. This can be confusing since CR eqs. do not account for the continuity of partial derivatives at every point in $f(z)$.

UNIT 5: (3) An arc that crosses over itself cannot be differentiable.

ANSWER: FALSE

FALSE: There are certain conditions that prove/disprove this statement, but for the sake of arcs it is generally false. In a simple arc, there is never any overlapping and $f(t_1) \neq f(t_2)$; $t_1 \neq t_2$ and t_1 and t_2 aren't endpoints. In a closed simple arc there is an extra condition where $z(a) = z(b)$. Therefore, it can be true that for an arc with some t_1 and t_2 , that $z(t_1) = z(t_2)$ holds true.

UNIT 6: (4) $\sum_{n=1}^{\infty} z_n$ converges iff $\lim_{N \rightarrow \infty} S_N = S$ given that $S_N = \sum_{n=1}^N z_n$.

ANSWER: TRUE

TRUE: The one condition the sum must fulfill in order to converge is that the sequence of partial sums must also converge as N approaches infinity. In other words, $\sum_{n=1}^{\infty} z_n = S$. This can be difficult since in this course we utilize real and imaginary values, as opposed to previously only real numbers. Therefore, oscillation may occur where we are not familiar and can throw us off, believing it to divergence instead.

UNIT 7: (5) It can be said $z_1 < z_2$ or $z_1 > z_2$ for some values of z .

ANSWER: FALSE

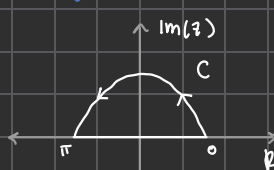
FALSE: Complex numbers can't be directly compared since unlike non-complex numbers, we must consider the imaginary parts as well. Complex numbers cannot be evaluated in one dimension, while real numbers can. However, we can say $|z_1| < |z_2|$ since this would measure the radius (r) of z , or distance from the origin, not value.

These questions relate to and emphasize this course as a whole because they give us a better understanding of domains in the complex plane and identifying if one is analytic, entire, or neither. As for the Cauchy-Riemann equations, we rely on heavily to determine differentiability or analyticity of a complex function, contour integrals, and mapping. The differentiability of arcs has an impact on how we utilize the Cauchy Integral Theorem and the inability to directly compare complex numbers emphasizes their analyticity and challenges our typical approach. And the convergence of a sum highlights sequences and series' place in approximation with the Taylor and Laurent expansions. All these things weigh heavily in complex variables and applications.

3 EXPLORE

Chapter 4, Section 46, #6

For the function f and contour C use parametric representations for C (legs of C) to evaluate $\int_C f(z) dz$ given. $f(z)$ is the principal branch $z^i = e^{i \log z}$; ($|z| > 0$, $-\pi < \text{Arg } z < \pi$) of the power function z^i , and C is the semicircle $z = e^{i\theta}$; ($0 \leq \theta \leq \pi$)



$$\begin{aligned}
 z(\theta) &= e^{i\theta} \rightarrow dz = ie^{i\theta} d\theta = [z'(\theta) = ie^{i\theta} d\theta] \\
 f(z) &= z^i = (e^{i\theta})^i = e^{-\theta} \rightarrow [f(z(\theta)) = e^{-\theta}] \\
 \int_C f(z) dz &= \int_0^\pi (e^{-\theta})(ie^{i\theta}) d\theta \\
 &= \int_0^\pi ie^{-\theta+i\theta} d\theta \\
 &= i \int_0^\pi e^{-\theta+i\theta} d\theta
 \end{aligned}$$

To evaluate the integral of a complex function f on a contour C , we parameterize C as $z(\theta) = x(\theta) + iy(\theta)$; $a \leq t \leq b$, and then

$$\int_C f(z) dz = \int_a^b f(z(\theta)) z'(\theta) d\theta$$

$$i \int_0^\pi e^{-\theta+i\theta} d\theta$$

$$\left. \frac{e^{-\theta+i\theta}}{i-1} \right|_0^\pi \rightarrow i \left[\frac{e^{-\pi+i\pi}}{i-1} - \frac{e^0}{i-1} \right]$$

$$\begin{aligned}
 e^{-\pi+i\pi} &= e^{-\pi} e^{i\pi} = e^{-\pi} (-1) = -e^{-\pi} \\
 e^{i\pi} &= \cos \pi + i \sin \pi \\
 &= -1 + i(0) \\
 &= -1
 \end{aligned}$$

to further simplify the expression to get it where we want, we must rationalize the denominator.

$$\begin{aligned}
 &\rightarrow i \cdot \frac{-e^{-\pi}-1}{i-1} \cdot \frac{(-i-1)}{(-i-1)} \\
 &= \frac{(-i e^{-\pi} - i)(-1-1)}{(i-1)(-i-1)} \\
 &= \frac{-e^{-\pi} + i e^{-\pi} - 1 + i}{1 - x^2 + 1} \\
 &= \frac{-1 - e^{-\pi} + i e^{-\pi} + i}{2} \\
 &= \frac{-1 - e^{-\pi}(1-i)}{2}
 \end{aligned}$$

Real:

$f(z) = z^i$
 reflects how f interacts with C
 providing a negative value

Imaginary:

$e^{-\pi}$ exponentially decreasing

contour integral:

$$\left[\frac{1 + e^{-\pi}(1-i)}{2} \right]$$

REFERENCES

- My own personal lecture / discussion notes and homework .
- "Class Notes" in files on catcourses
- Complex Variables and Applications textbook
- Discussion / Homework solutions in files on catcourses